Improved upper bound for the Turán density of daisies

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Abstract

For a graph F, the r-suspension $\hat{F}^r$ of F is the r-uniform hypergraph obtained by adjoining a fixed set S of $r-2$ new vertices to every edge of F. The (r,t)-daisy $\mathcal{D}_{r,t}$, introduced by Bollobás–Leader–Malvenuto, is the r-suspension of the complete graph K_t on t vertices.

A recent surprising result of Ellis–Ivan–Leader, disproving conjectures of Bollobás–Leader–Malvenuto and of Bukh, shows that the Turán density of $\mathcal{D}_{r,t+1}$ satisfies $\pi(\mathcal{D}_{r,t+1}) \geq 1 - 1/t - O(1/t^{1.475})$. We prove the complementary upper bound $\pi(\mathcal{D}_{r,t+1}) \leq 1 - 1/t - \Omega(1/t^8)$, showing that there is a genuine gap between $\pi(\mathcal{D}_{r,t+1})$ and $1-1/t$, the trivial upper bound given by Turán's theorem. Our proof uses the classical Birkhoff–von Neumann theorem on doubly stochastic matrices and an elegant result of Füredi on K_{t+1}-free graphs.

As a corollary, we obtain what appears to be the first nontrivial upper bound for a question of Füredi (Oberwolfach 2004) on the maximum edge density of 3-graphs whose link of every vertex is t-colorable. Another consequence of our result is that $\pi(\hat{F}^r)$ is strictly smaller than $\pi(F)$ for every non-bipartite graph F and every $r \geq 3$.

1 Introduction

Given an integer $r \geq 2$, an *r-uniform hypergraph* (henceforth an *r-graph*) $\mathcal{H}$ is a collection of r-subsets of some set V. We call V the *vertex set* of $\mathcal{H}$ and denote it by $V(\mathcal{H})$. When V is understood, we usually identify a hypergraph $\mathcal{H}$ with its edge set and use $|\mathcal{H}|$ to denote the number of edges of $\mathcal{H}$.

Given a family $\mathcal{F}$ of r-graphs, an r-graph $\mathcal{H}$ is *$\mathcal{F}$-free* if it does not contain any member of $\mathcal{F}$ as a subgraph. The *Turán number* $\mathrm{ex}(n,\mathcal{F})$ of $\mathcal{F}$ is the maximum number of edges in an $\mathcal{F}$-free r-graph on n vertices, and the limit $\pi(\mathcal{F}) := \lim_{n\to\infty} \mathrm{ex}(n,\mathcal{F})/\binom{n}{r}$ is called the *Turán density* of $\mathcal{F}$.

The study of Turán numbers and densities is a central topic in extremal combinatorics. While the case $r=2$ is well-understood due to the seminal work of Turán [Tur41] and Erdős–Stone [ES46] (see also [ES66]), determining $\pi(\mathcal{F})$ for $r \geq 3$ is notoriously difficult in general, despite significant effort devoted to this area. For a comprehensive overview of results up to 2011, we refer the reader to the excellent survey by Keevash [Kee11].

In this note, we investigate the Turán problem for a class of r-graphs closely related to extremal problems in hypercubes (see e.g. [Kos76, JE89, AKS07, JT10, Alo24, EIL25]) and posets (see e.g. [Buk09, GL16]). For a graph F, the *r-suspension* $\hat{F}^r$ of F is the r-graph obtained by adjoining a fixed set S of $r-2$ new vertices to every edge of F. The *(r,t)-daisy* $\mathcal{D}_{r,t}$, introduced by Bollobás–Leader–Malvenuto [BLM11], is the r-suspension of the complete graph K_t on t vertices.

It was conjectured by Bollobás–Leader–Malvenuto [BLM11, Conjecture 4] (and independently by Bukh for $t=4$) that for every fixed $t \geq 4$, $\lim_{r\to\infty} \pi(\mathcal{D}_{r,t}) = 0$. Notably, a simple averaging argument shows that for fixed t, the value $\pi(\mathcal{D}_{r,t})$ is non-increasing in r, and thus the limit is well-defined. Surprisingly, recent work by Ellis–Ivan–Leader [EIL24] disproved this conjecture, establishing the following lower bound:

Theorem 1.1 ([EIL24, Theorem 5])**.** *Suppose that $t-1$ is a prime power. Then*

$$\lim_{r\to\infty} \pi(\mathcal{D}_{r,t+1}) \geq \prod_{i=1}^{\infty}\left(1-\frac{1}{(t-1)^i}\right) = 1-\frac{1}{t}-\frac{2+o(1)}{t^2}.$$

In particular, there exists a constant $C>0$ such that for every $t\geq 3$,

$$\lim_{r\to\infty} \pi(\mathcal{D}_{r,t+1}) \geq 1-\frac{1}{t}-\frac{C}{t^{1.475}}.$$

The exponent 1.475 stems from a result of Baker–Harman–Pintz [BHP01], which shows that for all sufficiently large n, there exists a prime number in the interval $[n-n^{0.525}, n]$. This exponent 1.475 can be improved to $2-o(1)$ assuming Cramér's conjecture [Cra36], which is a major open problem in number theory.

We focus here on the upper bound of $\pi(\mathcal{D}_{r,t+1})$. Since $\pi(\mathcal{D}_{r,t+1})$ is non-increasing in r, the celebrated Turán Theorem implies that $\pi(\mathcal{D}_{r,t+1}) \leq 1-1/t$ for every $r\geq 3$. A natural question is whether a gap exists between $\pi(\mathcal{D}_{r,t+1})$ and $1-1/t$. Our main result confirms that such a gap indeed exists.

Theorem 1.2. *For every $r\geq 3$ and $t\geq 2$, we have*

$$\pi(\mathcal{D}_{r,t+1}) \leq 1-\frac{1}{t}-\frac{1}{30t^8}.$$

Determining the exact value of $\pi(\mathcal{D}_{r,t+1})$ for $r\geq 3$ appears to be rather difficult. Even the seemingly simplest case $(r,t)=(3,2)$ has been an open problem for decades (see e.g., [dC83, FF84, Mub03, BT11]). In our proof, we did not attempt to optimize the exponent in the upper bound above, and instead leave the determination of the growth rate of the gap $|\pi(\mathcal{D}_{r,t+1})-(t-1)/t|$ as an open problem for future research (see Section 4).

Below we state some consequences of Theorem 1.2.

For an r-graph $\mathcal{H}$ on vertex set V, the *link* of a vertex $v\in V$ in $\mathcal{H}$ is defined as

$$L_{\mathcal{H}}(v) := \left\{e\in \tbinom{V\setminus\{v\}}{r-1} \colon e\cup\{v\}\in\mathcal{H}\right\}.$$

An old conjecture of Erdős–Sós (see e.g. [FF84, p. 328]) states that if an n-vertex 3-graph $\mathcal{H}$ has the property that the link of every vertex is bipartite, then $|\mathcal{H}|\leq \frac{1}{4}\binom{n}{3}+o(n^3)$. This conjecture remains open, and one possible difficulty comes from the fact that if it is true, then there are many very different constructions achieving the bound asymptotically (see [FF84, MPS11, FRV13] for further discussion). Füredi (Oberwolfach 2004, see [MPS11, Problem 14]) considered the following generalization of the bipartite-link conjecture of Erdős–Sós.

Problem 1.3 (Füredi, Oberwolfach 2004)**.** *Let $t\geq 2$ be an integer. What is the (asymptotic) maximum size of an n-vertex 3-graph $\mathcal{H}$ where the link of every vertex is t-partite?*

Note that every hypergraph satisfying the conditions of Problem 1.3 is necessarily $\mathcal{D}_{3,t+1}$-free. Thus, we obtain the following upper bound, which appears to be the first nontrivial upper bound for Füredi's question.

Corollary 1.4. *Let $t\geq 2$ be an integer. Suppose that $\mathcal{H}$ is an n-vertex 3-graph in which the link of every vertex is t-partite. Then*

$$|\mathcal{H}| \leq \left(1-\frac{1}{t}-\frac{1}{30t^8}\right)\binom{n}{3}+o(n^3).$$

A construction based on recursive blowups of complements of projective planes, due to John Goldwasser (see [MPS11, p.11]), shows that the upper bound[1] cannot be smaller than

$$\frac{(t-1)^2}{t^2-t+2} = 1-\frac{1}{t}-\frac{(2+o(1))}{t^2}$$

[1] In fact, Goldwasser's construction is for $\mathcal{D}_{3,t+1}$-free 3-graphs; however, it satisfies the stronger property that the link of every vertex is t-partite.

when $t-1$ is a prime power. Consequently, by the result of Baker–Harman–Pintz [BHP01], the bound cannot be smaller than $1-1/t-O\left(1/t^{1.475}\right)$ for general t.

We remark that a minor modification of the proof of Theorem 1.2 yields an improved upper bound in Corollary 1.4 of $\left(1-\frac{1}{t}-\frac{c}{t^4}\right)\binom{n}{3}+o(n^3)$ for some constant $c>0$. Since improving the exponent of the error term c/t^4 further to 2 likely requires new ideas, we omit the proof here.

Finally, we address the Turán density of graph suspensions. It follows from the well-known Kövari–Sós–Turán Theorem [KST54] that every bipartite graph F satisfies $\pi(F)=0$. The analogous result for r-partite r-graphs was established by Erdős [Erd64]. Therefore, $\pi(\hat{F}^r)=\pi(F)=0$ if F is a bipartite graph. The following corollary of Theorem 1.2 shows that, for graphs, this is the only case in which equality holds, thereby settling a special case of [HLZ+25, Problem 4.2(i)].

Corollary 1.5. *Suppose that F is a non-bipartite graph. Then $\pi(\hat{F}^r)<\pi(F)$ for every $r\geq 3$.*

The remainder of this note is organized as follows. Section 2 introduces the necessary definitions and preliminary lemmas. Section 3 contains the proof of Theorem 1.2. Section 4 concludes with several remarks.

2 Preliminaries

In this section, we introduce the necessary definitions and preliminary lemmas.

Let G be a graph. For a set $S\subseteq V(G)$, we write $G[S]$ for the *induced subgraph* of G on S. For pairwise disjoint sets $V_1,\dots,V_t\subseteq V(G)$, we write $G[V_1,\dots,V_t]$ for the *induced t-partite subgraph* of G with parts $V_1,\dots,V_t$. For convenience, we write $K[V_1,\dots,V_t]$ for the complete t-partite graph with parts $V_1,\dots,V_t$. Thus, $G[V_1,\dots,V_t]=G\cap K[V_1,\dots,V_t]$.

For an r-graph $\mathcal{H}$, the *degree* of a vertex v in $\mathcal{H}$ is defined by $d_{\mathcal{H}}(v):=|L_{\mathcal{H}}(v)|$. We use $\delta(\mathcal{H})$, $\Delta(\mathcal{H})$, and $d(\mathcal{H})$ to denote the *minimum*, *maximum*, and *average* degrees of $\mathcal{H}$, respectively. The subscript $\mathcal{H}$ will be omitted when it is clear from the context.

Given two r-graphs F_1 and F_2, a map $\psi\colon V(F_1)\to V(F_2)$ is called a *homomorphism* from F_1 to F_2 if $\psi(e)\in F_2$ for every $e\in F_1$. We will use the following standard result in Turán-type problems, which follows from a supersaturation theorem of Erdős–Simonovits [ES83] (see also [Kee11, Theorem 2.2]).

Lemma 2.1. *Let F_1 and F_2 be r-graphs. If there exists a homomorphism from F_1 to F_2, then*

$$\pi(F_1)\leq\pi(F_2).$$

Recall that the *chromatic number* $\chi(F)$ of a graph F is the smallest positive integer t for which there exists a homomorphism from F to K_t. An immediate consequence of Lemma 2.1 is the following.

Fact 2.2. *Suppose that F is a graph with chromatic number $\chi(F)=t+1$. Then*

$$\pi(\hat{F}^r)\leq\pi(\mathcal{D}_{r,t+1}).$$

Thus, Corollary 1.5 follows directly from Fact 2.2 and the celebrated Erdős–Stone–Simonovits theorem [ES46, ES66], which states that, for every graph F,

$$\pi(F)=1-\frac{1}{\chi(F)-1}.$$

Let F be an r-graph. An r-graph $\mathcal{H}$ is called *F-hom-free* if there is no homomorphism from F to $\mathcal{H}$. We say that F is *blowup-invariant* if every F-free r-graph is also F-hom-free. Observe that, since every pair of vertices in $\mathcal{D}_{r,t+1}$ is contained in an edge, the r-graph $\mathcal{D}_{r,t+1}$ is blowup-invariant. Consequently, a standard deleting-duplicating argument (see, for example, [BLLP25, Proposition 2.6]) yields the following property of extremal $\mathcal{D}_{r,t+1}$-free hypergraphs.

Lemma 2.3. *Let $r \geq 2$ and $t \geq 2$ be integers. Suppose that $\mathcal{H}$ is a $\mathcal{D}_{r,t+1}$-free r-graph on n vertices with exactly* $\mathrm{ex}(n, \mathcal{D}_{r,t+1})$ *edges. Then*

$$\Delta(\mathcal{H}) - \delta(\mathcal{H}) \leq \binom{n-2}{r-2}.$$

For integers $n \geq t \geq 2$, let $T_{n,t}$ denote the balanced complete t-partite graph on n vertices.

A key ingredient in our proof is the following elegant result of Füredi [Für15, Theorem 2] on K_{t+1}-free graphs.

Theorem 2.4 ([Für15, Theorem 2])**.** *Let $n \geq t \geq 2$ and $p \geq 0$ be integers. Suppose that G is an n-vertex K_{t+1}-free graph with $|T_{n,t}| - p$ edges. Then G can be made t-partite by deleting at most p edges.*

Let $M \in \mathbb{R}^{n\times n}$ be an $n \times n$ matrix. A *transversal* of M is a set of n entries containing exactly one entry from each row and each column of M. We say that M is *doubly stochastic* if all its entries are nonnegative and every row sum and column sum equals one.

The classical Birkhoff–von Neumann theorem [Bir46] (see also [JDM60]), which can be derived from a standard application of Hall's theorem [Hal35], asserts that every $n \times n$ doubly stochastic matrix can be expressed as a convex combination of at most $(n-1)^2 + 1$ permutation matrices. A simple averaging argument therefore yields the following lemma, which will also be crucial for our proof.

Lemma 2.5. *Let $M \in \mathbb{R}^{n\times n}$ be a doubly stochastic matrix. Then there exists a transversal of M such that every entry in the transversal is at least $\left((n-1)^2 + 1\right)^{-1}$.*

3 Proof of Theorem 1.2

In this section, we present the proof of Theorem 1.2.

Proof of Theorem 1.2. Fix an integer $t \geq 2$, and let n be sufficiently large integer. Since $\pi(\mathcal{D}_{r,t+1})$ is nonincreasing in r, it suffices to show that $\pi(\mathcal{D}_{3,t+1}) \leq 1 - \frac{1}{t} - \frac{1}{30t^8}$.

Let $\mathcal{H}$ be a $\mathcal{D}_{3,t+1}$-free 3-graph on n vertices with exactly $\mathrm{ex}(n, \mathcal{D}_{3,t+1})$ edges, that is, $\mathcal{H}$ is an extremal 3-graph. Suppose, for the sake of contradiction, that $|\mathcal{H}| \geq \left(1 - \frac{1}{t} - \frac{1}{30t^8}\right)\binom{n}{3}$. By Lemma 2.3, we have

$$\delta(\mathcal{H}) \geq \Delta(\mathcal{H}) - (n-2) \geq d(\mathcal{H}) - n = \frac{3|\mathcal{H}|}{n} - n \geq \frac{1}{2}\left(1 - \frac{1}{t} - \frac{1}{29t^8}\right)n^2. \tag{1}$$

Let $V := V(\mathcal{H})$. For every vertex $x \in V$, fix a partition $V_1^x \cup \cdots \cup V_t^x = V \setminus \{x\}$ such that the size of the t-partite subgraph $L(x)[V_1^x, \ldots, V_t^x]$ of the link $L(x)$ is maximized. Define the set of bad and missing edges as follows

$$B^x := L(x) \setminus L(x)[V_1^x, \ldots, V_t^x] \quad \text{and} \quad M^x := K[V_1^x, \ldots, V_t^x] \setminus L(x).$$

Define also

$$Z^x := \left\{v \in V : d_{M^x}(v) \geq \frac{n}{10t^4}\right\}.$$

Claim 3.1. *For every $x \in V$, we have*

$$|B^x| \leq \frac{n^2}{58t^8}, \quad |M^x| \leq \frac{n^2}{29t^8}, \quad \text{and} \quad |Z^x| \leq \frac{20n}{29t^4}.$$

Proof of Claim 3.1. Fix $x \in V$. Since $\mathcal{H}$ is $\mathcal{D}_{3,t+1}$-free, the link $L(x)$ is K_{t+1}-free. Thus, by Theorem 2.4, $L(x)$ can be made t-partite by deleting at most $|T_{n,t}| - |L(x)|$ edges. Combining this with the maximality of $L(x)[V_1^x, \ldots, V_t^x]$ and (1), we obtain

$$|B^x| = |L(x)[V_1^x]| + \cdots + |L(x)[V_1^x]| \leq \frac{t-1}{2t}n^2 - \frac{1}{2}\left(1 - \frac{1}{t} - \frac{1}{29t^8}\right)n^2 = \frac{n^2}{58t^8}.$$

Consequently,

$$|M^x| = |K[V_1^x, \dots, V_t^x]| - |L(x)[V_1^x, \dots, V_t^x]| \le |T_{n,t}| - (|L(x)| - |B^x|) \le \frac{n^2}{58t^8} + \frac{n^2}{58t^8} = \frac{n^2}{29t^8}. \quad (2)$$

It remains to prove the upper bound for Z^x. By the definition of Z^x, we have

$$|M^x| = \frac{1}{2} \sum_{v \in V} d_{M^x}(v) \ge \frac{1}{2} \cdot |Z^x| \cdot \frac{1}{10t^4} n.$$

Combining this with (2), we obtain $|Z^x| \le \frac{20t^4|M^x|}{n} \le \frac{20n}{29t^4}$. ∎

Claim 3.2. *For every $x \in V$ and every $i \in [t]$, we have $\left||V_i^x| - \frac{n}{t}\right| \le \frac{n}{3t^4}$.*

Proof of Claim 3.2. Fix $x \in V$. By symmetry, it suffices to consider the case $i = 1$. Let $\alpha := |V_1^x|/n$. First, note that we have the lower bound

$$|L(x)[V_1^x, \dots, V_t^x]| = |L(x)| - |B^x| \ge \frac{1}{2}\left(1 - \frac{1}{t} - \frac{1}{29t^8}\right) n^2 - \frac{1}{58t^8} n^2 = \frac{1}{2}\left(1 - \frac{1}{t}\right) n^2 - \frac{1}{29t^8} n^2. \quad (3)$$

On the other hand, we have the upper bound

$$\begin{aligned} |L(x)[V_1^x, \dots, V_t^x]| &= \sum_{1 \le i < j \le t} |V_i^x||V_j^x| \\ &= \alpha n \cdot (n - \alpha n) + \sum_{2 \le i < j \le t} |V_i^x||V_j^x| \\ &\le \alpha(1-\alpha)n^2 + \binom{t-1}{2}\left(\frac{n - \alpha n}{t-1}\right)^2 = \left(\frac{-t}{2(t-1)}\alpha^2 + \frac{1}{t-1}\alpha + \frac{t-2}{2(t-1)}\right) n^2. \end{aligned}$$

Combining this with (3), we obtain

$$\frac{1}{2}\left(1 - \frac{1}{t}\right) - \frac{1}{29t^8} \le \frac{-t}{2(t-1)}\alpha^2 + \frac{1}{t-1}\alpha + \frac{t-2}{2(t-1)},$$

which implies that

$$(t\alpha - 1)^2 \le \frac{2(t-1)}{29t^7} \le \frac{1}{9t^6}.$$

It follows that $\left|\alpha - \frac{1}{t}\right| \le \frac{1}{3t^4}$, which proves Claim 3.2. ∎

Claim 3.3. *For every distinct pair $\{x, y\} \subseteq V$, there exists a permutation σ (depending on $\{x, y\}$) of the indices $\{1, \dots, t\}$ such that for every $i \in [t]$,*

$$\left|V_i^x \cap V_{\sigma(i)}^y\right| \ge \left(\frac{1}{t^3} - \frac{2}{3t^4}\right) n.$$

Proof of Claim 3.3. Fix a distinct pair $\{x, y\} \subseteq V$. Let $m := \left\lceil \frac{n}{t} + \frac{n}{3t^4} \right\rceil$, noting from Claim 3.2 that

$$\max\{|V_i^x| \colon i \in [t]\} \le m \quad \text{and} \quad \max\{|V_i^y| \colon i \in [t]\} \le m.$$

Let S be a set of $tm - n$ new vertices, disjoint from V. Let $\hat{V}_1^x \cup \dots \cup \hat{V}_t^x = V \cup S$ and $\hat{V}_1^y \cup \dots \cup \hat{V}_t^y = V \cup S$ be partitions such that

$$|\hat{V}_1^x| = \dots = |\hat{V}_t^x| = |\hat{V}_1^y| = \dots = |\hat{V}_t^y| = m \quad \text{and} \quad V_i^x \subseteq \hat{V}_i^x,\ V_i^y \subseteq \hat{V}_i^y \quad \text{for} \quad i \in [t].$$

Define a matrix $A = (a_{i,j})_{(i,j) \in [t]^2} \in \mathbb{R}^{t \times t}$ with the (i, j)-th entry given by

$$a_{i,j} := \left|\hat{V}_i^x \cap \hat{V}_j^y\right| / m \ge 0.$$

For every $i \in [t]$, we have

$$\sum_{j\in[t]} a_{i,j} = \sum_{j\in[t]} \left|\hat{V}_i^x \cap \hat{V}_j^y\right|/m = \left|\hat{V}_i^x\right|/m = 1.$$

By symmetry, for every $j \in [t]$, we have $\sum_{i\in[t]} a_{i,j} = 1$ as well. So the matrix A is a doubly stochastic matrix. Thus, it follows from Lemma 2.5 that there exists a permutation σ of $\{1,\dots,t\}$ such that for every $i \in [t]$,

$$a_{i,\sigma(i)} \geq \frac{1}{(t-1)^2+1} \geq \frac{1}{t^2}.$$

Consequently, for every $i \in [t]$,

$$\left|\hat{V}_i^x \cap \hat{V}_{\sigma(i)}^y\right| = a_{i,\sigma(i)} \cdot m \geq \frac{1}{t^2}\left(\frac{n}{t} + \frac{n}{3t^4}\right) = \left(\frac{1}{t^3} + \frac{1}{3t^6}\right)n. \tag{4}$$

Note that

$$\begin{aligned}\left|\hat{V}_i^x \cap \hat{V}_{\sigma(i)}^y\right| &= \left|V_i^x \cap V_{\sigma(i)}^y\right| + \left|\left(\hat{V}_i^x \setminus V_i^x\right) \cap \hat{V}_{\sigma(i)}^y\right| + \left|V_i^x \cap \left(\hat{V}_{\sigma(i)}^y \setminus V_{\sigma(i)}^y\right)\right| \\ &\leq \left|V_i^x \cap V_{\sigma(i)}^y\right| + \left|\hat{V}_i^x \setminus V_i^x\right| + \left|V_i^x \cap \left(\hat{V}_{\sigma(i)}^y \setminus V_{\sigma(i)}^y\right)\right| \leq \left|V_i^x \cap V_{\sigma(i)}^y\right| + \left|\hat{V}_i^x \setminus V_i^x\right| + 1,\end{aligned}$$

where the last inequality follows from the fact that there is at most one vertex, namely y, in the intersection $V_i^x \cap \left(\hat{V}_{\sigma(i)}^y \setminus V_{\sigma(i)}^y\right)$. Combining this with (4) and Claim 3.2, we obtain

$$\begin{aligned}\left|V_i^x \cap V_{\sigma(i)}^y\right| &\geq \left|\hat{V}_i^x \cap \hat{V}_{\sigma(i)}^y\right| - \left|\hat{V}_i^x \setminus V_i^x\right| - 1 \\ &\geq \left|\hat{V}_i^x \cap \hat{V}_{\sigma(i)}^y\right| - \left(m - \left(\tfrac{1}{t} - \tfrac{1}{3t^4}\right)n\right) - 1 \\ &\geq \left(\tfrac{1}{t^3} + \tfrac{1}{3t^6}\right)n - \left(\tfrac{1}{t} + \tfrac{1}{3t^4}\right)n + \left(\tfrac{1}{t} - \tfrac{1}{3t^4}\right)n - 2 \geq \left(\tfrac{1}{t^3} - \tfrac{2}{3t^4}\right)n,\end{aligned}$$

where the last inequality holds because n is large. This completes the proof of Claim 3.3. ∎

Fix a vertex $x \in V$. For every $y \in V \setminus \{x\}$, let σ_y be the permutation given by Claim 3.3. By relabeling the indices of $\{V_1^y, \dots, V_t^y\}$ if necessary, we may assume that $\sigma_y(i) = i$ for every $i \in [t]$, that is,

$$\left|V_i^x \cap V_i^y\right| \geq \left(\tfrac{1}{t^3} - \tfrac{2}{3t^4}\right)n \quad \text{for} \quad i \in [t]. \tag{5}$$

Claim 3.4. *Let $i \in [t]$ and $y \in V_i^x \setminus Z^x$. For every $j \in [t] \setminus \{i\}$, we have*

$$\left|V_j^x \cap V_i^y\right| \leq \frac{n}{6t^4}.$$

Proof of Claim 3.4. Fix $i \in [t]$ and $y \in V_i^x \setminus Z^x$. By symmetry, we may assume that $i = 1$. First, it follows from Claims 3.1 and 3.2 that

$$|V_1^x \setminus Z^x| \geq |V_1^x| - |Z^x| \geq \left(\frac{1}{t} - \frac{1}{3t^4}\right)n - \frac{20}{29t^4}n \geq \left(\frac{1}{t} - \frac{2}{t^4}\right)n \geq \left(\frac{1}{t} - \frac{1}{t^3}\right)n > 0.$$

Suppose for contradiction that there exists $j_0 \in [t] \setminus \{1\}$ such that $|V_{j_0}^x \cap V_1^y| \geq \frac{n}{6t^4}$. Choose a vertex z_1 uniformly at random from $V_{j_0}^x \cap V_1^y$. Independently, for each $k \in [2,t]$, choose a vertex z_k uniformly at random from $V_k^x \cap V_k^y$. For each $k \in [t]$, let X_k denote the event that $\{y, z_k\} \in L(x)$. For each $\{i,j\} \in \binom{[t]}{2}$, let $Y_{i,j}$ denote the event that $\{z_i, z_j\} \in L(y)$. We now calculate the probability that each event occurs.

Recall that $y \in V_1^x \setminus Z^x$ and $j_0 \in [t] \setminus \{1\}$. It follows from the definition of Z^x that

$$\mathbb{P}\left[\overline{X_1}\right] = \mathbb{P}\left[\{y, z_1\} \notin L(x)\right] \leq \frac{d_{M^x}(y)}{|V_{j_0}^x \cap V_1^y|} \leq \left(\tfrac{n}{10t^4}\right) / \left(\tfrac{n}{6t^4}\right) = \frac{3}{5}.$$

Fix $k \in [2,t]$. By (5), we have

$$\mathbb{P}\left[\overline{X_k}\right] = \mathbb{P}\left[\{y, z_k\} \notin L(x)\right] \leq \frac{d_{M^x}(y)}{|V_k^x \cap V_k^y|} \leq \left(\tfrac{n}{10t^4}\right) / \left(\tfrac{n}{t^3} - \tfrac{2n}{3t^4}\right) = \frac{3}{30t - 20}.$$

For every $j \in [2,t]$, it follows from (2) and (5) that

$$\mathbb{P}\left[\overline{Y_{1,j}}\right] = \mathbb{P}\left[\{z_1, z_j\} \notin L(y)\right] \le \frac{|M_y|}{|V_{j_0}^x \cap V_1^y| \cdot |V_j^x \cap V_j^y|} \le \left(\tfrac{n^2}{29t^8}\right) / \left(\tfrac{n}{6t^4}\left(\tfrac{n}{t^3} - \tfrac{2n}{3t^4}\right)\right) = \frac{18}{87t-58}.$$

Similarly, for every $\{i,j\} \in \binom{[2,t]}{2}$,

$$\mathbb{P}\left[\overline{Y_{i,j}}\right] = \mathbb{P}\left[\{z_i, z_j\} \notin L(y)\right] \le \frac{|M_y|}{|V_i^x \cap V_i^y| \cdot |V_j^x \cap V_j^y|} \le \left(\tfrac{n^2}{29t^8}\right) / \left(\tfrac{n}{t^3} - \tfrac{2n}{3t^4}\right)^2 = \frac{9}{29(3t-2)^2}.$$

Therefore, by the union bound, the probability that all events $\{X_k : k \in [t]\}$ and $\left\{Y_{i,j} : \{i,j\} \in \binom{[t]}{2}\right\}$ occur simultaneously satisfies

$$\mathbb{P}\Big[\bigwedge_{k\in[t]} X_k \wedge \bigwedge_{\{i,j\}\in\binom{[t]}{2}} Y_{i,j}\Big] \ge 1 - \frac{3}{5} - \frac{3(t-1)}{30t-20} - \frac{18(t-1)}{87t-58} - \frac{9\binom{t-1}{2}}{29(3t-2)^2} = \frac{99t^2+39t-80}{145(2-3t)^2} > 0.$$

Thus, there exists a selection of $\{z_1, \ldots, z_t\}$ such that all events $\{X_k : k \in [t]\}$ and $\left\{Y_{i,j} : \{i,j\} \in \binom{[t]}{2}\right\}$ occur simultaneously, that is,

$$\mathcal{G} \coloneqq \{yxz_i : i \in [t]\} \cup \left\{yz_iz_j : \{i,j\} \in \binom{[t]}{2}\right\} \subseteq \mathcal{H}.$$

However, note that $\mathcal{G}$ is a copy of $\mathcal{D}_{3,t+1}$ with y as its center, contradicting the assumption that $\mathcal{H}$ is $\mathcal{D}_{3,t+1}$-free. This completes the proof of the Claim 3.4. ∎

Define an auxiliary bipartite graph G with parts $U \coloneqq V_1^x$ and $W \coloneqq V_2^x \setminus Z^x$, where a pair $(u,w) \in U \times W$ forms an edge in G if and only if $u \notin V_2^w \cup Z^w$.

Claim 3.5. *There exists a vertex $u_0 \in U$ such that $d_G(u_0) \ge \left(\frac{1}{t} - \frac{1112}{435t^4}\right)n$.*

Proof of Claim 3.5. For every vertex $w \in W$, it follows from Claims 3.1, 3.2, and 3.4 that

$$d_G(w) \ge |V_1^x| - |V_1^x \cap V_2^w| - |Z^w| \ge \left(\frac{1}{t} - \frac{1}{3t^4}\right)n - \frac{20n}{29t^4} - \frac{n}{6t^4} \ge \left(\frac{1}{t} - \frac{6}{5t^4}\right)n.$$

Thus

$$\begin{aligned}|G| = \sum_{w\in W} d_G(w) \ge |W| \cdot \left(\frac{1}{t} - \frac{6}{5t^4}\right)n &\ge \left(\frac{n}{t} - \frac{n}{3t^4} - \frac{20n}{29t^4}\right) \cdot \left(\frac{1}{t} - \frac{6}{5t^4}\right)n \\ &= \left(\frac{1}{t^2} - \frac{967}{435t^5} + \frac{178}{145t^8}\right)n \ge \left(\frac{1}{t^2} - \frac{967}{435t^5}\right)n^2.\end{aligned}$$

By averaging, there exists a vertex $u_0 \in U$ such that

$$d_G(u_0) \ge \frac{|G|}{|U|} \ge \frac{\left(\frac{1}{t^2} - \frac{967}{435t^5}\right)n^2}{\left(\frac{1}{t} + \frac{1}{3t^4}\right)n} = \left(\frac{1}{t} - \frac{1112}{435t^4} + \frac{1112}{435t^4\,(3t^3+1)}\right)n \ge \left(\frac{1}{t} - \frac{1112}{435t^4}\right)n,$$

proving Claim 3.5. ∎

Note from Claims 3.2 and 3.4 that for every $w \in W$,

$$|V_2^w \cap V_2^x| = |V_2^w| - \sum_{j\in[t]\setminus\{2\}} \left|V_2^w \cap V_j^x\right| \ge \left(\frac{1}{t} - \frac{1}{3t^4}\right)n - (t-1)\cdot\frac{n}{6t^4} = \left(\frac{1}{t} - \frac{t+1}{6t^4}\right)n.$$

By the definition of G, for every $w \in N_G(u_0)$, we have $u_0 \notin V_2^w \cup Z^w$. Therefore, by the definition of Z^w,

$$\left|N_{L(w)}(u_0) \cap V_2^x\right| \ge |V_2^w \cap V_2^x| - d_{M^w}(u_0) \ge \left(\frac{1}{t} - \frac{t+1}{6t^4}\right)n - \frac{n}{10t^4} \ge \left(\frac{1}{t} - \frac{3}{10t^3}\right)n, \tag{6}$$

where the last inequality holds since $t \ge 2$.

Observe that for every vertex $v \in N_{L(w)}(u_0) \cap V_2^x$, we have $\{v, w\} \in L(u_0)[V_2^x]$. Combining this with Claim 3.5 and (6), we obtain

$$\begin{aligned} |L(u_0)[V_2^x]| &\geq \tfrac{1}{2} \cdot |N_G(u_0)| \cdot \left|N_{L(w)}(u_0) \cap V_2^x\right| \\ &\geq \tfrac{1}{2} \cdot \left(\tfrac{1}{t} - \tfrac{1112}{435t^4}\right) n \cdot \left(\tfrac{1}{t} - \tfrac{3}{10t^3}\right) n \\ &= \tfrac{1}{2}\left(\tfrac{1}{t^2} - \tfrac{3}{10t^4} - \tfrac{1112}{435t^5} + \tfrac{556}{725t^7}\right) n^2 \geq \tfrac{1}{2}\left(\tfrac{1}{t^2} - \tfrac{3}{10 \cdot 2t^3} - \tfrac{1112}{435 \cdot 4t^3}\right) n^2 = \tfrac{1}{2}\left(\tfrac{1}{t^2} - \tfrac{1373}{1740t^3}\right) n^2 \end{aligned} \tag{7}$$

On the other hand, since $L(u_0)$ is K_{t+1}-free, it follows from Turán's theorem and Claim 3.2 that

$$\begin{aligned} |L(u_0)[V_2^x]| &\leq \tfrac{1}{2}\left(1 - \tfrac{1}{t}\right) |V_2^x|^2 \leq \tfrac{1}{2}\left(1 - \tfrac{1}{t}\right)\left(\tfrac{1}{t} + \tfrac{1}{3t^4}\right)^2 n^2 \\ &= \tfrac{1}{2}\left(1 - \tfrac{1}{t}\right)\left(\tfrac{1}{t^2} + \tfrac{2}{3t^5} + \tfrac{1}{9t^8}\right) n^2 \\ &< \tfrac{1}{2}\left(\tfrac{1}{t^2} - \tfrac{1}{t^3} + \tfrac{2}{3t^5} + \tfrac{1}{9t^8}\right) n^2 \\ &< \tfrac{1}{2}\left(\tfrac{1}{t^2} - \tfrac{1}{t^3} + \tfrac{2}{3 \cdot 4t^3} + \tfrac{1}{9 \cdot 2^5 t^3}\right) n^2 = \tfrac{1}{2}\left(\tfrac{1}{t^2} - \tfrac{239}{288t^3}\right) n^2 < \tfrac{1}{2}\left(\tfrac{1}{t^2} - \tfrac{1373}{1740t^3}\right) n^2, \end{aligned}$$

a contradiction to (7). This completes the proof of Theorem 1.2. ∎

4 Concluding Remarks

In light of the lower bound established by Ellis–Ivan–Leader [EIL24] and the upper bound provided by Theorem 1.2, an immediate open question concerns the asymptotic behavior of the gap between the Turán density of $\mathcal{D}_{r,t+1}$ and the classical Turán bound $1 - 1/t$.

Problem 4.1. *For every integer $r \geq 3$, determine the growth rate of*

$$\left|\pi(\mathcal{D}_{r,t+1}) - \tfrac{t-1}{t}\right| \quad \text{as} \quad t \to \infty.$$

Building on the results in this note and the constructions of Goldwasser, we propose the following conjecture for Füredi's t-partite link problem, generalizing the Erdős–Sós bipartite link conjecture.

Conjecture 4.2. *Let $t \geq 2$ be an integer. Suppose that $\mathcal{H}$ is an n-vertex 3-graph in which the link of every vertex is t-partite. Then*

$$|\mathcal{H}| \leq \frac{(t-1)^2}{t^2 - t + 2}\binom{n}{3} + o(n^3).$$

We highlight an even more ambitious conjecture due to Goldwasser (see [MPS11, p.11]), which suggests that the density of $\mathcal{D}_{3,t+1}$ is governed by the same value.

Conjecture 4.3 (Goldwasser [MPS11, p.11])**.** *For every integer $t \geq 3$,*

$$\pi(\mathcal{D}_{3,t+1}) \leq \frac{(t-1)^2}{t^2 - t + 2}.$$

Finally, we remark on a phenomenon for suspensions that differs from the graph Turán problems. By the celebrated Erdős–Stone–Simonovits theorem [ES46, ES66], we have

$$\pi(F) = \pi(K_{\chi(F)}) = 1 - \frac{1}{\chi(F) - 1}.$$

However, for $r \geq 3$, the Turán density of $\hat{F}^r$ is not necessarily equal to that of $\mathcal{D}_{r,\chi(F)}$, the r-suspension of the complete graph $K_{\chi(F)}$.

For example, consider the 5-cycle C_5. There exists a homomorphism from $\hat{C}_5^3$ to $\mathcal{D}_{3,3}$ (i.e. K_4^{3-}). Using computer-assisted flag algebra calculations, Falgas-Ravry–Vaughan [FRV13, Theorem 3.19] showed that

$$\pi(\hat{C}_5^3) = \pi(K_4^{3-}, \hat{C}_5^3) \leq 0.258295\,,$$

while the construction of Frankl–Füredi [FF84] shows that

$$\pi(K_4^{3-}) \geq \frac{2}{7} = 0.285714... > \pi(\hat{C}_5^3).$$

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